Form B

Core Focus on Proportions & Probability

Block 1 Review ~ Ratios and Rates

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. Complete the conversion:
3 kilometers = _____ meters

A. 0.3
B. 30
C. 300
D. 3,000

D (1 km = 1,000 m, so 3 × 1,000 = 3,000 m)

2. Which numbers are equivalent to $3\frac{1}{3}$? Circle all that apply.

A. $3.1\overline{3}$
B. $3.\overline{3}$
C. $\frac{10}{3}$
D. $3\frac{2}{6}$
E. 3.3
F. $\frac{13}{3}$

B, C, D (1 ÷ 3 = 0.333…; 3 × 3 + 1 = 10, so 10/3; 2/6 = 1/3)

3. Eighteen of 24 dogs at an animal shelter were adopted this month. What is the ratio of dogs that were adopted to dogs that were not adopted?

A. 3 : 4
B. 3 : 1
C. 1 : 3
D. 3 : 7

B (24 − 18 = 6 not adopted; 18 : 6 = 3 : 1)

4. A class has a boys to girls ratio of 3 : 5. What is the ratio of girls to total students in the class?

A. 5 : 3
B. 3 : 8
C. 5 : 8
D. 8 : 5

C (total parts = 3 + 5 = 8; girls : total = 5 : 8)

5. Maya biked 27 miles in 1.5 hours. What was her speed in miles per hour?

A. 16 miles per hour
B. 18 miles per hour
C. 18.5 miles per hour
D. 40.5 miles per hour

B (27 ÷ 1.5 = 18 mph)

6. Which rate should be used to convert $\frac{48 \text{ inches}}{1 \text{ minute}}$ to $\frac{\text{feet}}{\text{minute}}$?

A. $\frac{12 \text{ inches}}{1 \text{ foot}}$
B. $\frac{1 \text{ foot}}{12 \text{ inches}}$
C. $\frac{60 \text{ seconds}}{1 \text{ minute}}$
D. $\frac{1 \text{ minute}}{60 \text{ seconds}}$

B (inches is on top, so put inches on the bottom: 48 ÷ 12 = 4 ft per min)

7. For numbers 7a – 7d, circle YES or NO to tell whether the rate given is equivalent to walking 2,400 meters per hour.

7a. 40 meters per minute	YES	NO
7b. 144,000 meters per minute	YES	NO
7c. 57,600 meters per day	YES	NO
7d. 40 meters per second	YES	NO

7a YES (2,400 ÷ 60 = 40) · 7b NO (that multiplies by 60) · 7c YES (2,400 × 24 = 57,600) · 7d NO (2,400 ÷ 3,600 = $\frac{2}{3}$ m per s)

8. What is the unit rate for $\frac{\frac{3}{4} \text{ mile}}{6 \text{ minutes}}$?

A. $4\frac{1}{2}$ miles per minute
B. 8 miles per minute
C. $\frac{1}{8}$ mile per minute
D. $\frac{3}{4}$ mile per minute

C ($\frac{3}{4} \div 6 = \frac{3}{4} \times \frac{1}{6} = \frac{3}{24} = \frac{1}{8}$)

9. Ari skated $2\frac{1}{4}$ miles in $1\frac{1}{2}$ hours. Assume he skated the entire time at the same rate. What was his speed in miles per hour?

A. $\frac{2}{3}$ mile per hour

B. $1\frac{1}{2}$ miles per hour

C. $3\frac{3}{8}$ miles per hour

D. $1\frac{1}{6}$ miles per hour

B ($\frac{9}{4} \div \frac{3}{2} = \frac{9}{4} \times \frac{2}{3} = \frac{18}{12} = 1\frac{1}{2}$ mph)

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Part II – Constructed Response

Complete each conversion.

1. 9 feet = _____ inches

108 inches (9 × 12)

2. 3 meters = _____ centimeters

300 centimeters (3 × 100)

3. 40 ounces = _____ pounds

2.5 pounds (40 ÷ 16)

4. 18 inches = _____ feet

1.5 feet (18 ÷ 12)

Convert each fraction to a decimal.

5. $\frac{3}{8}$

0.375 (3 ÷ 8)

6. $\frac{5}{9}$

$0.\overline{5}$ (5 ÷ 9 = 0.555...)

Convert each decimal to a fraction.

7. 0.6

$\frac{3}{5}$ ($\frac{6}{10} = \frac{3}{5}$)

8. 0.35

$\frac{7}{20}$ ($\frac{35}{100} \div 5 = \frac{7}{20}$)

9. Simplify the ratio 16 : 24.

2 : 3 (divide both by 8)

10. Fifteen of 20 students in the school band practiced over the weekend. Write three different simplified number ratios using this information. Explain the meaning of each ratio using words.

Ratio (numbers):	Ratio (numbers):	Ratio (numbers):
Explanation: 3 : 4 — students who practiced to all students in the band (15 : 20)	Explanation: 1 : 4 — students who did not practice to all students (5 : 20)	Explanation: 3 : 1 — students who practiced to students who did not (15 : 5). The reversed ratios 4 : 3, 4 : 1 and 1 : 3 are also correct if explained.

Find the unit rate.

11. $\frac{96 \text{ kilometers}}{4 \text{ hours}}$

24 kilometers per hour (96 ÷ 4)

12. $\frac{\frac{5}{8} \text{ meter}}{\frac{3}{4} \text{ second}}$

$\frac{5}{6}$ meter per second ($\frac{5}{8} \times \frac{4}{3} = \frac{20}{24} = \frac{5}{6}$)

13. Rosa spent $54.00 on six tickets to a basketball game. Dev spent $44 on four tickets. Which person paid less per ticket? Show all work necessary to justify your answer.

Rosa: $54 ÷ 6 = $9.00 per ticket. Dev: $44 ÷ 4 = $11.00 per ticket. Rosa paid less per ticket.

14. A cyclist rides at a rate of 15 miles per hour. Write this rate as feet per hour. Show all work necessary to justify your answer.

79,200 feet per hour ($\frac{15 \text{ mi}}{1 \text{ hr}} \times \frac{5{,}280 \text{ ft}}{1 \text{ mi}} = 79{,}200$ ft per hr)

15. A ferry travels at a rate of 18 kilometers per hour. Write this rate as meters per minute. Show all work necessary to justify your answer.

300 meters per minute ($\frac{18 \text{ km}}{1 \text{ hr}} \times \frac{1{,}000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ hr}}{60 \text{ min}} = \frac{18{,}000 \text{ m}}{60 \text{ min}} = 300$ m per min)

16. It took $\frac{3}{4}$ hour to wash 6 cars at a fundraiser. Assume each car took the same amount of time. What was the rate in hours per car? Nora answered this question, but made a mistake. Explain Nora's mistake. Then complete the work correctly to answer the question.

I need to find: $\frac{\frac{3}{4} \text{ hour}}{6 \text{ cars}}$

$\frac{\frac{3}{4} \text{ hour}}{6 \text{ cars}} = \frac{3}{4} \div \frac{1}{6}$

$= \frac{3}{4} \cdot \frac{6}{1}$

$= \frac{18}{4}$

$= 4\frac{1}{2}$ hours per car

Mistake: 6 cars is the whole number $\frac{6}{1}$, not $\frac{1}{6}$, so she divided by the wrong number (and $4\frac{1}{2}$ hours per car is more than the whole job took). Correct: $\frac{3}{4} \div 6 = \frac{3}{4} \times \frac{1}{6} = \frac{3}{24} = \frac{1}{8}$ hour per car.

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Part III – Problem Solving

There are 10 girls and 15 boys on the swim team. They swim laps in a swim-a-thon to raise money for new team suits. Each girl raised $6.00 per lap and swam at a rate of 12 laps per hour. Each boy raised $4.00 per lap and swam at a rate of 2 laps every 15 minutes. What is the ratio of the amount of money the girls raised in one hour to the amount of money the boys raised in one hour?

Girls: 10 girls × 12 laps × $6 = $720 per hour. Boys: 2 laps per 15 min = 8 laps per hour (60 ÷ 15 = 4, 2 × 4 = 8); 15 boys × 8 laps × $4 = $480 per hour. Ratio 720 : 480 = 3 : 2 (divide both by 240).